# Practical - 11

## Student Management CRUD with Search and Filter

**Problem Definition:** Create a student management module to add, view, edit, delete, search and filter student records. All database operations must use MYSQLi statements and clear success/failure messages.

## A. Theory

**1. Are SELECT, INSERT, UPDATE, and DELETE operations working? Yes — all four CRUD operations are implemented:**

SELECT → students.php fetches and displays all students from the students table.
INSERT → add_student.php adds a new student with duplicate check and password hashing.
UPDATE → edit_student.php pre-fills the form using SELECT, then updates the record on submit.
DELETE → students.php handles ?delete=id in the URL, runs DELETE FROM students WHERE student_id = $id.

**2. Is search by name/email/course implemented?**

Yes — the search form on students.php uses a LIKE '%search%' query on full_name and email, plus a dropdown filter for course. Both can be used together at the same time.

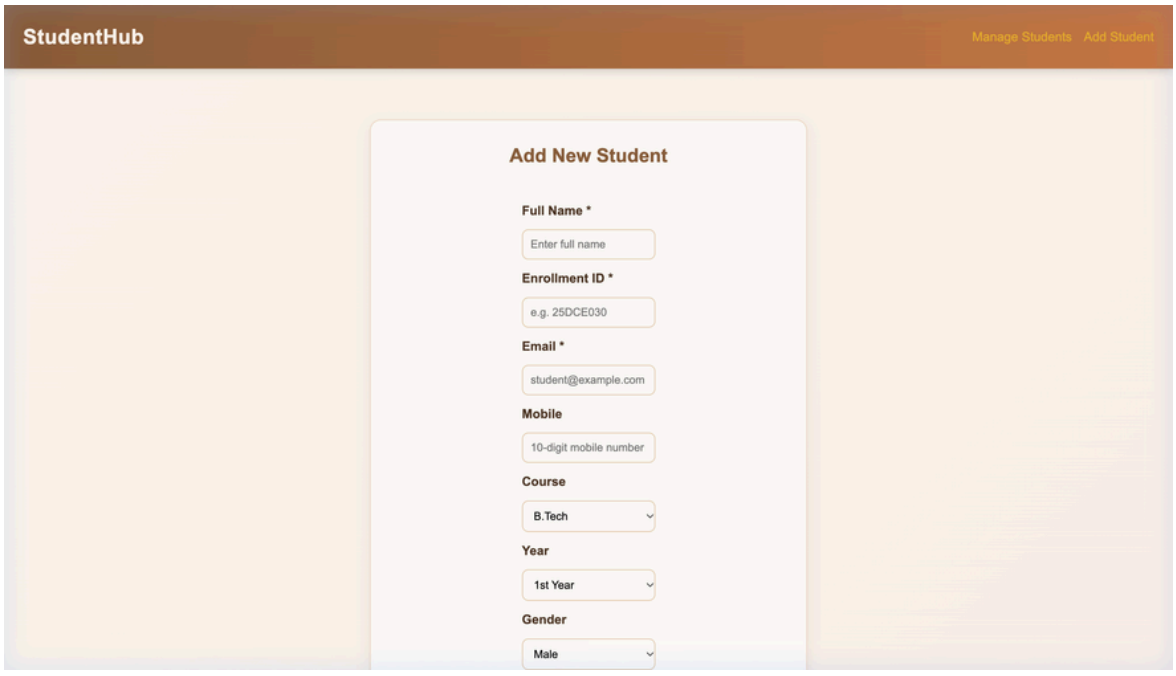
**3. Are delete operations confirmed before execution?**

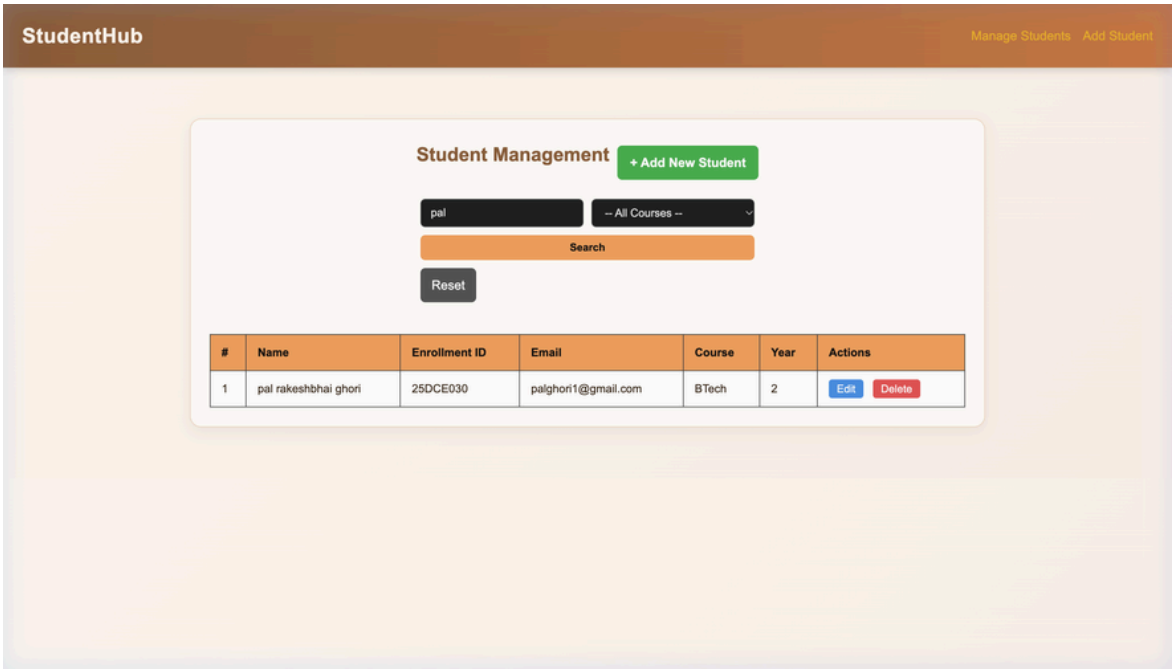
Yes — the Delete button uses JavaScript's built-in onclick="return confirm('Are you sure you want to delete this student?')". If the user clicks Cancel, the deletion does not happen.

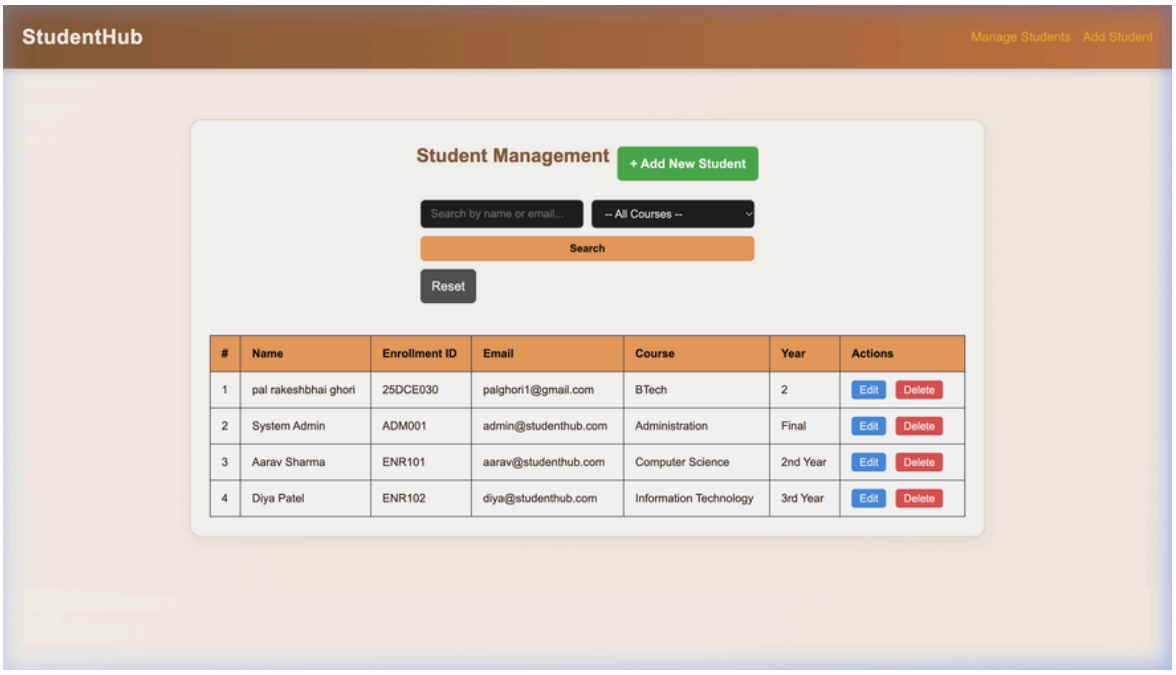
## B. Implementation

| File | What It Does |
|---|---|
| db_connect.php | Reused from Practical 9 — basic MySQLi |
| students.php | Lists all students, handles |
| add_student.php | Form to INSERT a new student |
| edit_student.php | Pre-fills form via SELECT, updates via UPDATE |

## C. Evidence of Work (Screenshots)




StudentHub
Manage Students
Add Student
Student Management
+ Add New Student
pal
-- All Courses --
Search
Reset
# Name Enrollment ID Email Course Year Actions
1 pal rakeshbhai ghori 25DCE030 palghori1@gmail.com BTech 2 Edit Delete


StudentHub
Manage Students
Add Student
Student Management
+ Add New Student
Search by name or email...
-- All Courses --
Search
Reset
# Name Enrollment ID Email Course Year Actions
1 pal rakeshbhai ghori 25DCE030 palghori1@gmail.com BTech 2 Edit Delete
2 System Admin ADM001 admin@studenthub.com Administration Final Edit Delete
3 Aarav Sharma ENR101 aarav@studenthub.com Computer Science 2nd Year Edit Delete
4 Diya Patel ENR102 diya@studenthub.com Information Technology 3rd Year Edit Delete

## D. GitHub Repository Link:

https://github.com/palghori/STUDENT-HUB-PORTAL-WDF/tree/main/practical%211

## E. Challenges Faced:

1. Building dynamic SQL queries — Adding WHERE 1=1 to the base query made it easy to optionally append search and filter conditions without breaking the syntax.
2. Pre-filling the edit form — Fetching the current row via SELECT and displaying its values as value="..." in the HTML form required careful use of htmlspecialchars() to avoid breaking the HTML if a student's name has quotes.
3. Preventing accidental deletes — A direct link without confirmation would delete data immediately, so onclick="return confirm(...)" was added as a safety net.

## F. Learning Outcomes:

1. Learned all four CRUD operations (Create, Read, Update, Delete) using MySQLi procedural style.
2. Understood how to build dynamic SQL queries with optional WHERE clauses for search and filter.
3. Learned how LIKE '%keyword%' works for pattern-based searching in MySQL.
4. Understood the importance of confirming destructive actions before executing them.
5. Learned how to pre-fill HTML forms from database data to create a smooth edit experience.

## Faculty Remarks: